

GRANGER CAUSAL PRIORITY AND CHOICE OF VARIABLES IN VECTOR AUTOREGRESSIONS

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Abstract—We derive a closed-form expression for the posterior probability of Granger noncausality in a gaussian vector autoregression with a conjugate prior. We also express in closed form the posterior probability of Granger causal priority, a more general relation that accounts for indirect effects between variables and therefore is suitable in a multivariate context. One can use these results to answer the classic question, “Is variable z relevant for variable x ?” and to choose variables for a vector autoregression.

I. Introduction

“Is variable z relevant for variable x ?” is a recurrent question in macroeconomics. In classic papers, Granger (1969) formalized this question, and Sims (1972) investigated whether money is relevant for (“Granger causes”) income. Later, the vector autoregression (VAR) became the dominant model in macroeconometrics due to Sims (1980), and Granger noncausality has probably been the most common restriction tested in VARs. While Sims (1972) focused on the bivariate relation between money and income, in the wake of the Great Recession, economists have been interested in the multivariate question as to which of the many financial variables are most relevant for macroeconomic variables like income and the price level. For example, are financial prices (e.g., interest rates on private debt) or financial quantities (e.g., monetary aggregates or credit aggregates) important for the macroeconomy once one controls for other variables?

This paper provides two analytical results for researchers wishing to test Granger-causal relations in Bayesian VARs, popular following Doan, Litterman, and Sims (1984). We work out a closed-form expression for the posterior probability of Granger noncausality in a gaussian VAR with a conjugate prior. Gaussianity and prior conjugacy are standard assumptions in Bayesian VARs. Furthermore, we express in closed form the posterior probability of Granger causal priority. The concept of Granger causal priority appears in unpublished work by Sims (2010) and Doan and Todd (2010), and it seems to be unfamiliar to most economists. Granger causal priority takes into account indirect effects between variables, and therefore it is more suitable in a multivariate context than Granger noncausality. We hope that this paper will popularize the concept of Granger causal priority.

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Using Granger causal priority, we investigate in a quarterly VAR which of a large number of macroeconomic and financial variables are relevant for income (real GDP), the price level, and a short-term interest rate controlled by monetary policy. Some traditionally prominent macroeconomic variables like the unemployment rate emerge as important, along with one class of financial variables: interest rates on private debt (e.g., the lending rate to nonfinancial corporations) and government debt. Monetary aggregates, credit aggregates, stock market variables, and exchange rates emerge as less important. The findings are similar in the United States and in the euro area, although in the euro area, there is some evidence that monetary aggregates and private sector credit aggregates can be useful.

The methodology that we develop can be used to answer the classic question of Granger and Sims, in a multivariate context, and quickly (because of the analytical results). We also highlight one specific use of the methodology: choosing variables for a VAR. Consider a researcher with an interest in a set of variables y_i (e.g., y_i can consist of income, the price level, and the short-term interest rate). The researcher has data on a set of variables y that contains the variables of interest, $y_i \subset y$, and also includes other variables, y_j , $y_j \equiv y \setminus y_i$. Let $y_j \subseteq y_j$ be a subset of the other variables. Can y_j be omitted from the VAR to forecast y_i ? Can y_j be omitted from the VAR to compute impulse responses of y_i to structural shocks? The answer to each question is linked to Granger causal priority. Suppose that y follows a VAR process. If y_i is Granger causally prior to y_j , the forecasts of y_i from a VAR with all variables y are equal to the forecasts of y_i from a smaller VAR that omits y_j . Thus, y_j can be omitted from the VAR to forecast y_i . Furthermore, if y_i is Granger causally prior to y_j and an additional condition holds, the impulse response of y_i to a structural shock of interest from a VAR with all variables y is equal to the impulse response of y_i to that shock from a smaller VAR that omits y_j . Thus, y_j can be omitted from the VAR to compute the impulse response of y_i to the structural shock of interest.

We evaluate the posterior probability that income, the price level, and the short-term interest rate are Granger causally prior to each other variable in our data set. This yields a ranking of all variables in the data set in terms of their relevance for the three variables of interest. We suggest that variables poorly placed in the ranking (those associated with high probabilities of Granger causal priority) can be omitted from a VAR with the variables of interest. In the literature, an assumption or an informal specification search determines which variables are chosen for a VAR. The choice based on the ranking that we suggest retains an element of informality, but it becomes possible to motivate a given

choice by referring to a statistic (the posterior probability of Granger causal priority) that is simple to compute and interpret. Furthermore, after specifying a loss function, it is also straightforward to choose variables formally (i.e., by minimizing the posterior expected loss as prescribed by Bayesian decision theory), and we show how to do this.

Currently, Bayesians wishing to test Granger noncausality have to rely on the frequentist likelihood ratio test or on Monte Carlo. The properties of the likelihood ratio test of a zero restriction in a Bayesian VAR with an informative prior are unclear. Zha (1999) studies Bayesian inference in a structural VAR with Granger-noncausality restrictions, but he does not consider testing such restrictions. A formal Bayesian test has been possible, though essentially unused in practice, by applying the Gibbs sampler of Waggoner and Zha (2003) to sample from the posterior density of the parameters of a VAR with a Granger-noncausality restriction and computing the marginal likelihood from the Gibbs output.

The literature initiated by George, Sun, and Ni (2008) studies Bayesian VARs with zero restrictions a priori independent across coefficients. We are concerned with zero restrictions that apply to appropriate subsets of coefficients. Furthermore, this literature aims at inference using a set of VARs with many different patterns of zero restrictions via model averaging.

The literature on fundamentalness started by Hansen and Sargent (1991) and Lippi and Reichlin (1993) studies if structural shocks are fundamental, that is, if one can obtain structural shocks from current and past values of data. This literature tells us that failure to include in a VAR a variable that Granger-causes the included variables is a sufficient condition for nonfundamentalness (see Giannone & Reichlin, 2006, and Forni & Gambetti, 2014). Testing for Granger causal priority in one's data set as we do identifies the variables that need to be included in order to eliminate this kind of nonfundamentalness.

Section II defines Granger causal priority and explains the relationship between it and the choice of variables. Section III derives a closed-form expression for the posterior probability of Granger noncausality in a gaussian VAR with a conjugate prior. Section IV shows how to evaluate the posterior probability of Granger causal priority. In section V, we apply this paper's methodological results to data. Section VI concludes. Online appendixes contain additional material.

II. Granger Causal Priority and Choice of Variables

In this section we define Granger causal priority. We also describe the relationship between it and the choice of variables.

Throughout the paper, we assume that the set of variables y follows a VAR:

$$y(t) = \gamma + B(L)y(t-1) + u(t), \quad (1)$$

where $y(t)$ denotes y in period $t = 1, \dots, T$, γ is a constant term, $B(L)$ is a matrix polynomial in the lag operator of order

$P-1$, $P \geq 1$, and $u(t)$ is a gaussian vector with mean 0 and variance-covariance matrix Σ conditional on $y(t-s)$ for all $s \geq 1$. We denote with N the number of variables in y .

We assume in this section that the parameters of this data-generating process are known, that is, the values of γ , $B(p)$ for all $p = 1, \dots, P$, and Σ are known. We consider inference in the subsequent sections, where we assume that a data set with $T+P$ observations of y is available.

A. Granger Noncausality and Granger Causal Priority

Let us begin by recalling the concept of Granger noncausality. Granger (1969) proposed that variable z is relevant for (i.e., causes) variable x if z helps predict x . He formalized this idea in terms of the variance of the prediction error one period ahead. We state Granger's definition in the way in which it has been used in the VAR literature:

Definition 1. *Granger noncausality: Consider y_i and y_j , non overlapping subvectors of y . In the VAR in equation (1), y_j does not Granger-cause y_i if the coefficients on all lags of y_j in the equations with y_i on the left-hand side are equal to 0, $B_{ij}(L) = 0$.*

The likelihood ratio test relying on an asymptotic χ^2 statistic is a well-known frequentist test of the restriction $B_{ij}(L) = 0$.¹ In section III, we show how a Bayesian econometrician can evaluate the posterior probability of this restriction in a simple way.

Let us turn to Granger causal priority. The next definition, which we think is unfamiliar to most economists, appears in unpublished work by Sims (2010) and Doan and Todd (2010):

Definition 2. *Granger causal priority: In the VAR in equation (1), y_i is Granger causally prior to y_j if it is possible to partition all the variables in y into two subsets, y_1 and y_2 , such that $y_i \subseteq y_1$, $y_j \subseteq y_2$, and y_2 does not Granger-cause y_1 .*

This definition states that y_i is Granger causally prior to y_j in the VAR in equation (1) if the VAR has the following recursive form:

$$\begin{aligned} y_i &\rightarrow \begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix} = \begin{pmatrix} \gamma_1 \\ \gamma_2 \end{pmatrix} + \begin{pmatrix} B_{11}(L) & B_{12}(L) \\ B_{21}(L) & B_{22}(L) \end{pmatrix} \\ &\quad \times \begin{pmatrix} y_1(t-1) \\ y_2(t-1) \end{pmatrix} + \begin{pmatrix} u_1(t) \\ u_2(t) \end{pmatrix} \\ &\quad \text{with } B_{12}(L) = 0. \end{aligned} \quad (2)$$

In the special case when y consists only of y_i and y_j , Granger causal priority is the same as Granger noncausality. If there are other variables in y in addition to y_i and y_j , the set of coefficients in $B_{ij}(L)$ is a strict subset of the set of coefficients

¹ See, e.g., Hamilton (1994, chap. 11).

in $B_{12}(L)$; that is, Granger causal priority requires a stronger restriction than Granger noncausality. Consider an example. Suppose that $y = \{x, w, z\}$, x , w , and z are scalars, $y_i = x$, $y_j = z$, and y follows the VAR:

$$\begin{pmatrix} x(t) \\ w(t) \\ z(t) \end{pmatrix} = \begin{pmatrix} B_{xx} & B_{xw} & B_{xz} \\ B_{wx} & B_{ww} & B_{wz} \\ B_{zx} & B_{zw} & B_{zz} \end{pmatrix} \begin{pmatrix} x(t-1) \\ w(t-1) \\ z(t-1) \end{pmatrix} + u(t).$$

If $B_{xz} = 0$, $x(t+1) = B_{xx}x(t) + B_{xw}w(t) + u_1(t+1)$, and thus z does not Granger-cause x . However, $x(t+2) = \dots + B_{xw}B_{wz}z(t) + \dots$, that is, the two-period-ahead forecast of x depends on the current value of z so long as $B_{xw}B_{wz} \neq 0$. If $B_{xw}B_{wz} \neq 0$, z helps predict x two periods ahead indirectly, through the effect of z on the third variable in the system, w . (This point is made by Lütkepohl, 1993). To account for indirect effects, Dufour and Renault (1998) refine Granger's definition by defining Granger noncausality at a horizon $h \geq 1$. In their terminology, z does not Granger-cause x at horizon $h = 2$ if $B_{xz} = 0$ and $B_{xw}B_{wz} = 0$. Dufour and Renault show that Granger causal priority, which they call "the separation condition," is a sufficient condition for Granger noncausality at all horizons. In our example, x is Granger causally prior to z if either $B_{xz} = B_{xw} = 0$ or $B_{xz} = B_{wz} = 0$. In either case, our example VAR becomes recursive as in equation (2) with $x \subseteq y_1$ and $z \subseteq y_2$. Dufour and Renault also find a necessary-and-sufficient condition for Granger noncausality at all horizons in a VAR of any dimension. This condition is complex and therefore difficult to test in practice.²

B. Choice of Variables

Granger causal priority has implications for choosing variables for a VAR.

Consider a researcher who wants to predict y_i . Suppose that y_i is Granger causally prior to y_j , that is, there exists an appropriate partition of y into y_1 and y_2 with $B_{12}(L) = 0$. Then the forecasts of y_i from the VAR in equation (1) are equal at all horizons to the forecasts of y_i from the VAR

$$y_1(t) = \gamma_1 + B_{11}(L)y_1(t-1) + u_1(t), \quad (3)$$

where the variance-covariance matrix of $u_1(t)$ is Σ_{11} , the appropriate submatrix of Σ . By "the forecasts are equal," we mean that for given parameters γ , $B(p)$ for all $p = 1, \dots, P$, and Σ and for given data $y(t-P+1), \dots, y(t)$, at any horizon $h \geq 1$ the predictive density of $y_i(t+h)$ implied by model (2) is equal to the predictive density of $y_i(t+h)$ implied by model (3). Consequently, any point forecasts are also equal.³ It follows that if y_i is Granger causally

prior to y_j , the researcher can omit y_j (as well as all other variables in y_2) from the VAR to forecast y_i and the forecasts of y_i do not change. To justify omitting y_j , we need Granger causal priority; Granger noncausality does not suffice. Granger noncausality— $B_{ij}(L) = 0$ in model (1)—does not imply that the forecasts of y_i from model (1) are equal to the forecasts of y_i from a smaller VAR that omits y_j except in the following two special cases: (a) if y consists only of y_i and y_j or (b) if we want to forecast y_i only one period ahead.

What if y_i is not Granger causally prior to y_j ? Granger causal priority of y_i to y_j is a sufficient condition for y_j not to affect the forecasts of y_i at any horizon; it is not a necessary condition. However, testing the necessary-and-sufficient condition is difficult, whereas we show that evaluating the posterior probability of Granger causal priority is straightforward. Taking all this into consideration, we think that a simple and prudent approach is to check if y_i is Granger causally prior to y_j and, if it is not, to include y_j in the VAR.

Granger causal priority also has implications for choosing variables when the goal is impulse response analysis rather than prediction. Let ε denote the structural shocks that generate the variation in y . Suppose that a researcher is interested in the impulse response of y_i to a subset of the structural shocks $\varepsilon_k \subset \varepsilon$. It turns out that if y_i is Granger causally prior to y_j and a particular zero restriction on the contemporaneous impulse response of y to ε holds, the impulse response of y_i to ε_k from model (1) is equal to the impulse response of y_i to ε_k from model (3). The researcher can omit y_j from the VAR to compute the impulse response of y_i to ε_k and the impulse response does not change. For a given identification scheme, one can check if the necessary restriction on the contemporaneous impulse response of y to ε holds. The details are in online appendix A.

III. Granger Noncausality: A Closed-Form Bayes Factor

We turn to inference. In this section, we consider a Bayesian econometrician who wants to evaluate the posterior odds in favor of a hypothesis of Granger noncausality. The posterior odds are equal to the prior odds multiplied by the Bayes factor in favor of the hypothesis. We derive a closed-form expression for the Bayes factor in favor of Granger noncausality in a gaussian VAR with a conjugate prior.

Suppose that y follows the VAR in equation (1). Let B be the $K \times N$ matrix of stacked coefficients, $B = (B(1), \dots, B(P), \gamma)'$, where $K = NP + 1$ is the number of right-hand-side variables in each equation in the VAR. The likelihood of the data conditional on initial observations is

$$p(Y|B, \Sigma) = (2\pi)^{-NT/2} |\Sigma|^{-T/2} \times \exp\left(-\frac{1}{2} \text{tr}(Y - XB)'(Y - XB)\Sigma^{-1}\right),$$

² In the frequentist context, Dufour, Pelletier, and Renault (2006) study Granger noncausality at $h > 1$ using auxiliary models, and Dufour and Taamouti (2010) use simulations.

³ With a natural prior, the posterior predictive density of $y_i(t+h)$ (i.e., the predictive density that incorporates the uncertainty about the parameters) is identical in models (2) and (3). See section III.

where

$$Y_{T \times N} = \begin{pmatrix} y(1)' \\ y(2)' \\ \vdots \\ y(T)' \end{pmatrix} \quad \text{and} \quad X_{T \times K} = \begin{pmatrix} y(0)' & y(-1)' & \dots & y(1-P)' & 1 \\ y(1)' & y(0)' & \dots & y(2-P)' & 1 \\ \vdots & \vdots & & \vdots & \vdots \\ y(T-1)' & y(T-2)' & \dots & y(T-P)' & 1 \end{pmatrix}.$$

Consider a zero restriction on a subset of coefficients in this VAR. The restriction has the form that in each affected equation, the coefficients of the same right-hand-side variables are restricted. Formally, let α denote the indexes of a subset of the equations. Let β denote the indexes of a subset of the right-hand-side variables. Consider the restriction

$$B_{\beta\alpha} = \mathbf{0}, \quad (4)$$

where $B_{\beta\alpha}$ denotes the matrix consisting of the intersection of rows β and columns α of the matrix B and $\mathbf{0}$ denotes the matrix of 0s of the same size as $B_{\beta\alpha}$. A Granger-noncausality restriction is a special case of restriction equation (4).⁴

We turn to the specification of the prior in the unrestricted VAR and the prior in the restricted VAR. The priors matter in the Bayesian test of a restriction. For example, if we begin with a prior that puts a lot of probability in the neighborhood of $B_{\beta\alpha} = \mathbf{0}$, the additional value of restriction (4) will be small. By contrast, if our prior about B is very loose, we are bound to prefer the restriction (the phenomenon known as Lindley's paradox). Therefore, we pay attention to the prior throughout the paper.

A. Prior in the Unrestricted VAR and the Restricted VAR

Let ω^U denote the unrestricted VAR. We assume that the prior density of B and Σ in the unrestricted VAR, $p(B, \Sigma|\omega^U)$, is proper and conjugate:

$$p(B, \Sigma|\omega^U) \propto |\Sigma|^{-(\tilde{v}+K+N+1)/2} \times \exp\left(-\frac{1}{2} \text{tr}(\tilde{Y} - \tilde{X}B)'(\tilde{Y} - \tilde{X}B)\Sigma^{-1}\right), \quad (5)$$

where $\tilde{v} > N - 1$, $\tilde{Y}$ and $\tilde{X}$ are hyperparameters of appropriate dimensions and $\tilde{X}'\tilde{X}$ and $\tilde{Y}'(I - \tilde{X}(\tilde{X}'\tilde{X})^{-1}\tilde{X}')\tilde{Y}$ are nonsingular. We emphasize that the standard prior in the Bayesian VAR literature, the prior of Sims and Zha (1998) consisting of a modified Minnesota prior and additional dummy observations, is a special case of prior (5).

⁴ One can use the following analytical Bayes factor also to test a restriction other than Granger noncausality so long as the restriction satisfies the condition given in this paragraph.

Expression (5) is a kernel of a normal-inverted-Wishart density. It is straightforward to show that

$$p(B, \Sigma|\omega^U) = p(B|\Sigma, \omega^U)p(\Sigma|\omega^U) = \mathcal{N}(\text{vec } \tilde{B}, \Sigma \otimes \tilde{Q}) \mathcal{IW}(\tilde{S}, \tilde{v}), \quad (6)$$

where $\mathcal{N}$ denotes the multivariate normal density and $\mathcal{IW}$ denotes the inverted Wishart density,

$$\tilde{B} = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'\tilde{Y}, \quad \tilde{Q} = (\tilde{X}'\tilde{X})^{-1}, \quad \text{and} \\ \tilde{S} = (\tilde{Y} - \tilde{X}\tilde{B})'(\tilde{Y} - \tilde{X}\tilde{B}).$$

Furthermore, the posterior density of B and Σ , $p(B, \Sigma|Y, \omega^U)$, satisfies

$$p(B, \Sigma|Y, \omega^U) = p(B|\Sigma, Y, \omega^U)p(\Sigma|Y, \omega^U) = \mathcal{N}(\text{vec } \bar{B}, \Sigma \otimes \bar{Q}) \mathcal{IW}(\bar{S}, \bar{v}), \quad (7)$$

where

$$\bar{B} = (\bar{X}'\bar{X})^{-1}\bar{X}'\bar{Y}, \quad \bar{Q} = (\bar{X}'\bar{X})^{-1}, \\ \bar{S} = (\bar{Y} - \bar{X}\bar{B})'(\bar{Y} - \bar{X}\bar{B}), \\ \bar{X} = \begin{pmatrix} \tilde{X} \\ X \end{pmatrix}, \quad \bar{Y} = \begin{pmatrix} \tilde{Y} \\ Y \end{pmatrix}, \quad \text{and} \quad \bar{v} = \tilde{v} + T.$$

Let ω^R denote the restricted VAR. Let $B_{(\beta\alpha)}$ denote the vector collecting all coefficients in B other than the coefficients in $B_{\beta\alpha}$. We assume that the prior density of $B_{(\beta\alpha)}$ and Σ in the restricted VAR, $p(B_{(\beta\alpha)}, \Sigma|\omega^R)$, satisfies

$$p(B_{(\beta\alpha)}, \Sigma|\omega^R) = p(B_{(\beta\alpha)}, \Sigma|\omega^U, B_{\beta\alpha} = \mathbf{0}). \quad (8)$$

Equation (8) states that the prior in the restricted model is equal to the prior in the unrestricted model conditional on the restriction. This is the most natural assumption one can make concerning the prior in model ω^R given a prior in model ω^U . Consider a researcher who holds prior $p(B, \Sigma|\omega^U)$. Suppose that the researcher learns that $B_{\beta\alpha} = \mathbf{0}$. Following the rules of probability, the researcher will update the prior precisely using equation (8). In addition to being intuitive, assumption (8) helps us derive a closed-form expression for the Bayes factor in favor of Granger noncausality. We do not see how one can derive a closed-form expression for this Bayes factor without assumption (8).⁵

Equation (8) states that the prior density in the restricted VAR, $p(B_{(\beta\alpha)}, \Sigma|\omega^R)$, is a conditional density of the normal-inverted-Wishart density $p(B, \Sigma|\omega^U)$. While $p(B_{(\beta\alpha)}, \Sigma|\omega^R)$ is not normal-inverted-Wishart, a marginal density of $p(B_{(\beta\alpha)}, \Sigma|\omega^R)$ can belong to the normal-inverted-Wishart family. In particular, the marginal density of the parameters of the equations indexed by α (i.e., the restricted equations) is

⁵ Assumption (8) enables us to use the Savage-Dickey density ratio, as we will explain. When assumption (8) fails to hold, the results of Verdinelli and Wasserman (1995) generalizing the Savage-Dickey density ratio may speed up the necessary numerical approximation of the Bayes factor.

normal-inverted-Wishart. Furthermore, with a standard prior, the posterior predictive density of y_1 in the restricted VAR is equal to the posterior predictive density of y_1 in model (3). See online appendix B for details.

B. Closed-Form Bayes Factor

Let $p(Y|\omega^U)$ denote the marginal likelihood of the unrestricted model ω^U . Let $p(Y|\omega^R)$ denote the marginal likelihood of the restricted model ω^R .

Proposition 1. *The Bayes factor in favor of model ω^R , defined in expressions (1), (4), and (8), relative to model ω^U , defined in expressions (1) and (5), is given by*

$$\begin{aligned} \frac{p(Y|\omega^R)}{p(Y|\omega^U)} &= \frac{\Gamma_{N_\alpha} \left(\frac{\bar{v}-N_{(\alpha)}+K_\beta}{2} \right)}{\Gamma_{N_\alpha} \left(\frac{\bar{v}-N_{(\alpha)}}{2} \right)} \frac{\Gamma_{N_\alpha} \left(\frac{\bar{v}-N_{(\alpha)}}{2} \right)}{\Gamma_{N_\alpha} \left(\frac{\bar{v}-N_{(\alpha)}+K_\beta}{2} \right)} \\ &\times \frac{|\bar{S}_{\alpha\alpha}|^{\frac{\bar{v}-N_{(\alpha)}}{2}} |\bar{Q}_{\beta\beta}|^{-\frac{N_\alpha}{2}}}{|\tilde{S}_{\alpha\alpha}|^{\frac{\bar{v}-N_{(\alpha)}}{2}} |\tilde{Q}_{\beta\beta}|^{-\frac{N_\alpha}{2}}} \\ &\times \frac{|\bar{S}_{\alpha\alpha} + \bar{B}'_{\beta\alpha} (\bar{Q}_{\beta\beta})^{-1} \bar{B}_{\beta\alpha}|^{-\frac{\bar{v}-N_{(\alpha)}+K_\beta}{2}}}{|\tilde{S}_{\alpha\alpha} + \tilde{B}'_{\beta\alpha} (\tilde{Q}_{\beta\beta})^{-1} \tilde{B}_{\beta\alpha}|^{-\frac{\bar{v}-N_{(\alpha)}}{2}}}, \quad (9) \end{aligned}$$

where N_α is the number of the restricted equations, $N_{(\alpha)}$ is the number of the unrestricted equations, K_β is the number of the right-hand-side variables whose coefficients are restricted, and $\Gamma_N(\cdot)$ denotes the multivariate gamma function of dimension N , $\Gamma_N(z) = \pi^{N(N-1)/4} \prod_{j=1}^N \Gamma(z + (1-j)/2)$.

Proof. *Step 1a:* Given equation (6) the marginal prior density of B is matricvariate Student implying that the marginal prior density of $B_{\beta\alpha}$ is also matricvariate Student.

Step 1b: Given equation (7) the marginal posterior density of B is matricvariate Student, implying that the marginal posterior density of $B_{\beta\alpha}$ is also matricvariate Student. Steps 1a and 1b, as well as the parameters of the two densities of $B_{\beta\alpha}$, follow directly from Bauwens, Lubrano, and Richard (1999, appendix A.2.7).

Step 2: The Savage-Dickey result of Dickey (1971) implies that if the prior in the restricted model ω^R satisfies condition (8), the Bayes factor in favor of the restricted model ω^R relative to the unrestricted model ω^U is equal to the ratio of the marginal posterior density of $B_{\beta\alpha}$ evaluated at $B_{\beta\alpha} = \mathbf{0}$ to the marginal prior density of $B_{\beta\alpha}$ evaluated at $B_{\beta\alpha} = \mathbf{0}$. Therefore, equation (9) is obtained as the ratio of the marginal posterior density of $B_{\beta\alpha}$ from step 1b evaluated at $B_{\beta\alpha} = \mathbf{0}$ to the marginal prior density of $B_{\beta\alpha}$ from step 1a evaluated at $B_{\beta\alpha} = \mathbf{0}$.

Using Bayes factor (9) one can evaluate the posterior odds in favor of the hypothesis of Granger noncausality or,

equivalently, the posterior probability of this hypothesis. A researcher who wants to evaluate the posterior odds can proceed as follows: (a) specify the prior odds in favor of model ω^R relative to model ω^U (it is common to specify the prior odds to be uninformative, i.e., the prior odds equal to 1); (b) use equation (9) to compute the Bayes factor in favor of ω^R relative to ω^U ; and (c) multiply the prior odds by the Bayes factor to obtain the posterior odds. The posterior odds in favor of ω^R relative to ω^U are the posterior odds in favor of the Granger-noncausality restriction. The posterior probability of ω^R conditional on ω^R and ω^U is a simple transformation of the posterior odds, $\pi/(1+\pi)$, where π denotes the posterior odds.⁶

Proposition 1 is of independent interest. In the next section, we use it to evaluate the posterior probability of Granger causal priority.

IV. Posterior Probability of Granger Causal Priority

In this section we derive a closed-form expression for the posterior probability that y_i is Granger causally prior to y_j .

In section III, we evaluated the posterior probability of Granger noncausality conditional on the set of models with two elements: the restricted model ω^R and the unrestricted model ω^U . By contrast, evaluating the posterior probability of Granger causal priority is complicated by the fact that there are multiple partitions of y consistent with Granger causal priority of y_i to y_j (in the realistic case when there are variables in y that belong to neither y_i nor y_j). In other words, there are multiple restricted models consistent with Granger causal priority of y_i to y_j . We propose to evaluate the posterior probability of Granger causal priority conditional on the set of models Ω .

Definition 3. *Let Ω be the set of models such that (a) each model in Ω is a VAR of the form given in equation (1), (b) Ω includes the unrestricted VAR, and (c) Ω includes all VARs with the restriction $B_{12}(L) = 0$ for some partition of y into two subsets, y_1 and y_2 , such that $y_i \subseteq y_1$. Furthermore, let Ω^j be the subset of Ω containing all models in which $y_j \subseteq y_2$, that is, all models in which y_i is Granger causally prior to y_j .*

We continue to assume as in section III that the prior in the unrestricted model in Ω satisfies expression (5) and the prior in each restricted model in Ω satisfies condition (8) for appropriate α and β . Furthermore, we assume that all models in Ω have equal prior probabilities. It is straightforward to allow for different prior probabilities.

Evaluating the posterior probability that y_i is Granger causally prior to y_j conditional on Ω is equivalent to evaluating the posterior probability of Ω^j conditional on Ω ,

⁶ Suppose that one wants to obtain the marginal likelihood of the restricted VAR. Since the marginal likelihood of an unrestricted gaussian VAR with a conjugate prior is available in closed form, one simply multiplies the Bayes factor given in equation (9) by the marginal likelihood of the unrestricted VAR.

$p(\Omega^j|Y, \Omega)$. Furthermore, $p(\Omega^j|Y, \Omega)$ can be expressed in closed form:

$$\begin{aligned} p(\Omega^j|Y, \Omega) &= \frac{p(\Omega^j|Y)}{p(\Omega|Y)} = \frac{\sum_{\omega_k \in \Omega^j} p(\omega_k|Y)}{\sum_{\omega_l \in \Omega} p(\omega_l|Y)} \\ &= \frac{\sum_{\omega_k \in \Omega^j} p(Y|\omega_k)p(\omega_k)/p(Y)}{\sum_{\omega_l \in \Omega} p(Y|\omega_l)p(\omega_l)/p(Y)} \\ &= \frac{\sum_{\omega_k \in \Omega^j} p(Y|\omega_k)}{\sum_{\omega_l \in \Omega} p(Y|\omega_l)} \\ &= \frac{\sum_{\omega_k \in \Omega^j} p(Y|\omega_k)/p(Y|\omega^U)}{\sum_{\omega_l \in \Omega} p(Y|\omega_l)/p(Y|\omega^U)}. \end{aligned} \quad (10)$$

The first equality follows from the definition of conditional probability and the fact that $\Omega^j \subset \Omega$. The second equality follows from the definitions of Ω and Ω^j . The third equality follows from Bayes' law. The fourth equality follows from the assumption that the prior probability $p(\omega)$ is equal for all models; thus, the terms $p(\omega_k)/p(Y)$ and $p(\omega_l)/p(Y)$ are all equal to one another. The fifth equality follows after we divide the numerator and the denominator by $p(Y|\omega^U)$. The final expression is a ratio of two sums of Bayes factors where each Bayes factor can be expressed in closed form by proposition 1.

If we think of “ y_i is Granger-causally-prior to y_j ” as the null hypothesis and we condition on Ω , the alternative hypothesis is the complement of Ω^j , $\Omega \setminus \Omega^j$, and $p(\Omega \setminus \Omega^j|Y, \Omega) = 1 - p(\Omega^j|Y, \Omega)$. Let N_j denote the number of variables in y_j . If y_j consists of a single variable (as in the application in the next section), 2^{N_j-1} models are consistent with the null (i.e., Ω^j has 2^{N_j-1} elements) and 2^{N_j-1} models are consistent with the alternative (i.e., $\Omega \setminus \Omega^j$ has 2^{N_j-1} elements). Hence the null and the alternative are symmetric.⁷

The main advantage of evaluating the posterior probability of Granger causal priority conditional on Ω is that in this case, the posterior probability can be expressed in closed form (because the marginal likelihood of each model in Ω can be written in closed form). The posterior probability of Granger causal priority conditional on a different set of models—for example, models that fail to satisfy equation (4), (5) or (8))—generally cannot be expressed in closed form.

⁷ In our example in section IIA, we have $N_j = 2$, and Ω consists of the following four models:

$$\begin{pmatrix} B_{xx} & B_{xw} & B_{xz} \\ B_{wx} & B_{ww} & B_{wz} \\ B_{zx} & B_{zw} & B_{zz} \end{pmatrix}, \begin{pmatrix} B_{xx} & 0 & 0 \\ B_{wx} & B_{ww} & B_{wz} \\ B_{zx} & B_{zw} & B_{zz} \end{pmatrix}, \begin{pmatrix} B_{xx} & B_{xw} & 0 \\ B_{wx} & B_{ww} & 0 \\ B_{zx} & B_{zw} & B_{zz} \end{pmatrix}, \begin{pmatrix} B_{xx} & 0 & B_{xz} \\ B_{wx} & B_{ww} & B_{wz} \\ B_{zx} & 0 & B_{zz} \end{pmatrix},$$

of which the second and the third belong to Ω^j , and the first and the fourth belong to $\Omega \setminus \Omega^j$. Online appendix C describes an approach in which the alternative hypothesis is represented by ω^U alone. That approach requires only a straightforward modification of the calculations shown here. Furthermore, in online appendix C, we explain why the approach in the paper seems preferable at least in some applications, including the application in this paper.

A computer may be too slow to calculate all sums in expression (10) when N_j is a large number (in practice, when N_j is greater than about 20). In the application in the next section, N_j is a large number ($N_j = 38$). Therefore, we approximate $p(\Omega^j|Y, \Omega)$ using the Markov chain Monte Carlo model composition algorithm of Madigan and York (1995). Online appendix D explains how we implement this algorithm and assess its convergence. Here we emphasize that this algorithm is simple and fast, and it converges reliably. It took us about six hours on a standard PC to produce reliable numerical approximations of all posterior probabilities, that is, $p(\Omega^j|Y, \Omega)$ for all $y_j \in y_J$, in our application.

V. Application

This section illustrates how to use the paper's methodological results.

A. Data

We put together two data sets, one for the euro area and one for the United States. Each data set has three variables of interest (i.e., in each data set, y_i has 3 elements) and 38 other variables (i.e., in each data set y_J has 38 elements). The variables of interest (the elements of y_i) are income (real GDP), a measure of the price level, and a short-term interest rate controlled by monetary policy. In the euro area exercise, we use the harmonized index of consumer prices (HICP) and the overnight interbank interest rate Eonia. In the U.S. exercise, we use the consumer price index (CPI) and the federal funds rate. In the euro area exercise, the other variables (i.e., the elements of y_J) are U.S. real GDP, the U.S. CPI, the federal funds rate, and 35 euro-area variables listed in the next paragraph. In the U.S. exercise, the other variables are euro-area real GDP, the HICP for the euro area, the Eonia, and 35 U.S. variables listed in the next paragraph.

The 35 variables are (a) components of real GDP (consumption, government consumption, investment, exports, imports, change in inventories); (b) labor market variables (unit labor cost, employment, unemployment rate, hours worked); (c) interest rates (two-year and ten-year government bond yields, the corporate bond spread defined as the spread between corporate bonds rated BBB with maturity seven to ten years and government bonds with the same maturity, lending rate to nonfinancial corporations, mortgage interest rate); (d) monetary aggregates (M1, M2, M3); (e) credit aggregates (government debt, loans for house purchase, consumer loans, loans to nonfinancial corporations); (f) exchange rates (nominal exchange rate between the U.S. dollar and the euro, nominal effective exchange rate); (g) commodity prices and other price indexes (price of oil, commodity prices, consumer price index excluding energy and food, producer price index); (h) housing market variables (house price index, real housing investment); (i) stock market variables (stock index, stock volatility

index); (j) survey-based indicators of economic activity (capacity utilization, purchasing managers' index); and (k) survey-based indicators of economic sentiment (industrial confidence, consumer confidence).⁸

The main sample contains quarterly data from 1999Q1 to 2013Q3. In the euro-area exercise, we decided to use data starting in 1999Q1 because this is when the monetary union began. We then decided to use the same period in the U.S. exercise for comparability. In addition, we used a training sample in each exercise.

B. Prior

The prior in the unrestricted model ($\omega^U \in \Omega$) consists of two pieces: (a) an initial prior formulated before seeing data and (b) a training sample prior. Formally, matrices $\tilde{Y}$ and $\tilde{X}$ in expression (5) have the form

$$\tilde{Y} = \begin{pmatrix} Y_{SZ} \\ Y_{ts} \end{pmatrix}, \quad \tilde{X} = \begin{pmatrix} X_{SZ} \\ X_{ts} \end{pmatrix}, \quad (11)$$

and $\tilde{v} = v_{SZ} + T_{ts}$, where Y_{SZ} , Y_{ts} , X_{SZ} , X_{ts} , v_{SZ} , and T_{ts} are defined below. Given this prior in the unrestricted model, the prior in each restricted model (in each $\omega^R \in \Omega$) is defined by equation (8). The initial prior is the prior of Sims and Zha (1998), the standard prior in Bayesian VARs, implemented via dummy observations Y_{SZ} and X_{SZ} . The prior of Sims and Zha (1998) treats variables symmetrically, and thus it implies no beliefs about which y_j 's are important for y_i and which y_j 's are unimportant. The training sample prior contains data from the sample 1989Q1 to 1998Q4 collected in matrices Y_{ts} and X_{ts} . T_{ts} is equal to the number of periods in the training sample minus the number of lags P .⁹ The prior of Sims and Zha (1998) is governed by v_{SZ} and five other hyperparameters. In the baseline specification, we use the hyperparameter values employed by Sims and Zha (1998) and standard in the literature, except that we set $v_{SZ} = N + 20$. (See online appendix F for the details.) In the baseline specification, we use four lags ($P = 4$), as is typical of a quarterly VAR in the literature. We report how our findings change as we move away from the baseline specification by varying the prior and the number of lags.

C. Probability of Granger Causal Priority: Baseline

Table 1 shows the posterior probability that y_i (real GDP, the price level, and the short-term interest rate) is Granger causally prior to a variable y_j , for each variable $y_j \in y_J$, in the euro area (left column) and in the United States (right column) in the baseline specification. We can now answer

the classic question: Is variable z relevant for variable x ? For example, we find that house prices are of little relevance for y_i . The probability that y_i is Granger causally prior to house prices is 0.95 in the euro area and 0.91 in the United States. As another example, we find that the dollar-euro exchange rate is of little relevance for y_i . The probability that y_i is Granger-causally-prior to the dollar-euro exchange rate is 1 in the euro area and in the United States.

The following variables are most relevant (the probability that y_i is Granger causally prior to a variable in this group is approximately 0): (a) survey-based indicators of economic activity and sentiment (industrial confidence and purchasing managers' index in the euro area, capacity utilization, industrial confidence, and purchasing managers' index in the United States); (b) interest rates on private debt and government debt (the mortgage interest rate, the lending rate to nonfinancial corporations, and the yield on two-year government bonds); (c) selected components of real GDP (the change in inventories, investment, exports, and imports); (d) the unemployment rate; and (e) the price of oil. This finding is similar for the euro area and for the United States. In general, the rank correlation between the posterior probabilities in the euro area and in United States is high: 0.6.

Which financial variables are relevant for macroeconomic variables? We find that interest rates (on both private debt and government debt) tend to be important for y_i , whereas monetary aggregates, credit aggregates, stock market variables, and exchange rates tend to be unimportant. This finding holds for the euro area and for the United States, although private sector credit aggregates are of some use in the euro area. Our interpretation is that interest rates respond quickly to news relevant for the macroeconomy. Monetary aggregates and credit aggregates react to news with delay and therefore contain little information not already present in other time series. Stock market variables and exchange rates respond quickly to news, but much of their variation is driven by information of little relevance for variables like real GDP and the price level.

The findings also let us choose variables for a VAR with y_i . Table 1 gives a ranking of variables in terms of their relevance for y_i . Researchers can justify omitting y_j from a VAR with y_i by reporting that y_j is poorly placed in the ranking. They can motivate the inclusion of y_j by stating that y_j turns up at the top of the ranking. A researcher who prefers a smaller VAR will choose fewer variables than someone who is comfortable with a larger model. Choosing variables in this manner retains an element of informality dominant in the literature today, but here it is possible to motivate a given choice by referring to a ranking simple to compute and interpret.

Since the marginal likelihood of a model in Ω is available in closed form, it is also straightforward to choose variables formally, following Bayesian decision theory. A formal choice requires specifying a loss function (one that assigns a numerical loss to every combination of actual choice of variables and the correct choice of variables) and minimizing

⁸ The details of all data are in online appendix E. Note that due to data availability, we include hours worked only for the United States, and we include M3 only for the euro area. The source of all data is the database of the ECB. The data are available from the authors on request.

⁹ Adding the training sample prior improves the marginal likelihood of the unrestricted model in the main sample compared with using only the prior of Sims and Zha (1998) in both the euro-area exercise and U.S. exercises.

TABLE 1.—POSTERIOR PROBABILITY THAT REAL GDP, PRICE LEVEL, AND SHORT-TERM INTEREST RATE ARE GRANGER CAUSALLY PRIOR TO A VARIABLE

Euro Area			United States	
Variable	Probability	Rank	Variable	Probability
Industrial confidence	0.00	1	Lending rate to NFCs	0.00
Mortgage interest rate	0.00	2	Capacity utilization	0.00
Real exports	0.00	3	Unemployment rate	0.00
Price of oil	0.00	4	Change in real inventories	0.00
Change in real inventories	0.00	5	Two-year government bond yield	0.00
Real investment	0.00	6	Real investment	0.00
Two-year government bond yield	0.00	7	Price of oil	0.00
Real imports	0.00	8	10-year government bond yield	0.00
Lending rate to NFCs	0.00	9	Industrial confidence	0.00
Employment	0.00	10	Mortgage interest rate	0.00
Real consumption	0.00	11	Purchasing managers' index	0.00
Unemployment rate	0.00	12	Real imports	0.00
Purchasing managers' index	0.00	13	Euro-area consumer prices	0.00
Producers price index	0.00	14	Producers price index	0.00
Loans to NFCs	0.00	15	Real exports	0.01
Loans for house purchase	0.01	16	Stock index	0.01
Consumer loans	0.02	17	Real consumption	0.01
Unit labor cost	0.06	18	Corporate bond spread	0.01
Real housing investment	0.09	19	Employment	0.02
Capacity utilization	0.11	20	M1	0.02
M3	0.24	21	Hours worked	0.03
M2	0.24	22	Consumer prices excl. energy, food	0.07
U.S. consumer prices	0.27	23	Nominal effective exchange rate	0.15
Ten-year government bond yield	0.33	24	Consumer confidence	0.15
M1	0.42	25	Commodity prices	0.25
Federal funds rate	0.49	26	Real government consumption	0.36
Real government consumption	0.74	27	Stock volatility index	0.39
Commodity prices	0.75	28	Unit labor cost	0.46
House prices	0.95	29	Loans to NFCs	0.48
U.S. real GDP	0.96	30	Euro area real GDP	0.53
Consumer prices excl. energy, food	0.96	31	Real housing investment	0.57
Consumer confidence	0.99	32	Loans for house purchase	0.59
Stock index	1.00	33	Government debt	0.64
Corporate bond spread	1.00	34	House prices	0.91
Nominal effective exchange rate	1.00	35	Consumer loans	1.00
Government debt	1.00	36	Dollar-euro exchange rate	1.00
Dollar-euro exchange rate	1.00	37	Eonia	1.00
Stock volatility index	1.00	38	M2	1.00

the posterior expected loss.¹⁰ We pursue this route in online appendix G where we assume a standard loss function, the 0-1 loss function. A researcher with this loss function selects the model implying the highest marginal likelihood of the data, $\omega^* \in \Omega$. Let y_1^* denote y_1 in this model, and let y_2^* denote y_2 in this model. It follows from section II that we can omit all variables in y_2^* from the VAR with y_i . As it turns out, we can omit variables from the bottom of the ranking in table 1. Thus, the formal choice of variables with the 0-1 loss function is consistent with the ranking.

We also computed root mean squared errors of out-of-sample point forecasts of the elements of y_i from a number of models. VARs including top-ranked variables did well in this exercise compared with a VAR including bottom-ranked variables, the VAR including all variables in the data set, the trivariate VAR with y_i only, and the random-walk-with-drift benchmark. (The details are in online appendix H).

¹⁰ By “the correct choice of variables,” we mean the choice one would make in the case when one knew if y_i is Granger causally prior to y_j , for each $y_j \in y_i$ (as opposed to having to infer from data if y_i is Granger causally prior to y_j).

D. Probability of Granger Causal Priority: Robustness

How do the findings change with the prior? We loosen or tighten the prior relative to the baseline specification by changing the value of a single scalar as in Giannone, Lenza, and Primiceri (2015). The prior density of B conditional on Σ implied by the prior of Sims and Zha (1998) in the baseline, can be written as $\mathcal{N}(\text{vec } B_{SZ}, \Sigma \otimes Q_{SZ})$, where $B_{SZ} = (X'_{SZ} X_{SZ})^{-1} X'_{SZ} Y_{SZ}$ and $Q_{SZ} = (X'_{SZ} X_{SZ})^{-1}$. We replace the term $\Sigma \otimes Q_{SZ}$ with $\xi \Sigma \otimes Q_{SZ}$ where ξ is a positive scalar. Now $\xi = 1$ is the baseline, whereas $\xi > 1$ implies loosening the prior and $\xi < 1$ implies tightening the prior relative to the baseline.

Table 2 (euro area) and table 3 (United States) report the results for $\xi = 4$ (“loosest”), $\xi = 2$ (“looser”), $\xi = 1/2$ (“tighter”), and $\xi = 1/4$ (“tightest”) in comparison with the baseline ($\xi = 1$). The looser the prior is, the higher the posterior probability of Granger causal priority (see the “average GCP probability” row). This is what we expected to find. If the prior is loose, the fit of a VAR is likely to improve significantly after we add a zero restriction. If the prior is tight, the fit of a VAR is likely to improve little after we add a zero

TABLE 2.—POSTERIOR PROBABILITY THAT REAL GDP, PRICE LEVEL, AND SHORT-TERM INTEREST RATE ARE GRANGER CAUSALLY PRIOR TO A VARIABLE, ROBUSTNESS, EURO AREA

Variable	Loosest $\xi = 4$	Looser $\xi = 2$	Baseline $\xi = 1$	Tighter $\xi = 1/2$	Tightest $\xi = 1/4$	Number of Lags, P			
						6	5	3	2
Industrial confidence	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Mortgage interest rate	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Real exports	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Price of oil	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Change in real inventories	0.99	0.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Real investment	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01
Two-year government bond yield	0.00	0.00	0.00	0.02	0.35	0.00	0.00	0.00	0.00
Real imports	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Lending rate to NFCs	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Employment	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.01
Real consumption	1.00	0.31	0.00	0.00	0.00	0.00	0.00	0.01	0.04
Unemployment rate	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Purchasing managers' index	0.01	0.00	0.00	0.01	0.01	0.00	0.00	0.00	0.04
Producers price index	0.00	0.00	0.00	0.03	0.04	0.00	0.00	0.06	0.26
Loans to NFCs	0.00	0.00	0.00	0.01	0.01	0.00	0.00	0.01	0.17
Loans for house purchase	0.00	0.01	0.01	0.01	0.01	0.01	0.01	0.08	0.70
Consumer loans	0.46	0.18	0.02	0.01	0.02	0.00	0.00	0.01	0.14
Unit labor cost	0.00	0.09	0.06	0.05	0.03	0.00	0.01	0.49	0.35
Real housing investment	0.54	0.15	0.09	0.02	0.02	0.00	0.01	0.13	0.46
Capacity utilization	0.00	0.01	0.11	0.67	0.52	0.01	0.04	0.15	0.40
M3	0.00	0.38	0.24	0.13	0.03	0.01	0.10	0.87	0.93
M2	0.00	0.38	0.24	0.13	0.03	0.00	0.10	0.87	0.93
U.S. consumer prices	0.02	0.27	0.27	0.45	0.34	0.04	0.13	0.61	0.74
Ten-year government bond yield	1.00	0.91	0.33	0.17	0.33	0.33	0.39	0.21	0.18
M1	0.00	0.10	0.42	0.69	0.69	0.01	0.03	0.96	0.95
Federal funds rate	0.95	0.44	0.49	0.04	0.08	0.03	0.06	0.36	0.15
Real government consumption	1.00	0.95	0.74	0.22	0.05	0.86	0.91	0.09	0.05
Commodity prices	0.91	0.74	0.75	0.83	0.54	0.30	0.60	0.91	0.75
House prices	0.00	0.01	0.95	1.00	0.98	0.72	0.89	0.98	1.00
U.S. real GDP	1.00	0.90	0.96	0.91	0.62	0.97	0.97	0.87	0.81
Consumer prices excluding energy, food	1.00	1.00	0.96	0.55	0.03	0.89	0.94	0.99	0.99
Consumer confidence	1.00	1.00	0.99	0.46	0.00	0.81	0.92	1.00	0.98
Stock index	1.00	1.00	1.00	0.99	0.89	1.00	1.00	1.00	0.99
Corporate bond spread	1.00	1.00	1.00	0.98	0.57	0.97	1.00	1.00	1.00
Nominal effective exchange rate	1.00	1.00	1.00	1.00	0.94	1.00	1.00	1.00	1.00
Government debt	1.00	1.00	1.00	1.00	0.99	1.00	1.00	1.00	1.00
Dollar-euro exchange rate	1.00	1.00	1.00	1.00	0.96	1.00	1.00	1.00	1.00
Stock volatility index	1.00	1.00	1.00	1.00	0.99	1.00	1.00	1.00	1.00
Average GCP probability	0.42	0.36	0.36	0.33	0.26	0.29	0.32	0.41	0.45
Rank correlation with the baseline	0.77	0.91	1.00	0.95	0.80	0.96	0.97	0.97	0.93
$\log p(Y \omega^*) - \log p(Y \omega^U)$	354	215	127	63	27	145	137	113	87
$\log p(Y \Omega)$	4,335	4,575	4,664	4,624	4,530	4,659	4,665	4,655	4,627

The column “baseline” reproduces the results from table 1. “GCP” stands for Granger causal priority. $\log p(Y|\omega^*) - \log p(Y|\omega^U)$ is the log Bayes factor in favor of the model with the highest marginal likelihood ($\omega^* \in \Omega$) relative to the unrestricted model ($\omega^U \in \Omega$). $p(Y|\Omega)$ is the marginal likelihood conditional on the set of models Ω .

restriction. The prior matters, as we anticipated in section III. To get another perspective, the penultimate row reports the log Bayes factor in favor of the model with the highest marginal likelihood ($\omega^* \in \Omega$) relative to the unrestricted model ($\omega^U \in \Omega$), $\log p(Y|\omega^*) - \log p(Y|\omega^U)$. The Bayes factor falls monotonically as the prior tightens, consistent with the intuition that Granger-noncausality restrictions become less valuable as the prior tightens.

The rank correlation between the posterior probabilities with the baseline prior and any other prior we consider is high (see the “rank correlation with the baseline” row). In other words, the ranking of the y_j ’s in terms of their relevance for y_i changes little as we vary the prior. Consequently, the answer to the question “Is z_1 more relevant for x than z_2 is?” is robust to the prior.

The findings concerning the role of the prior suggest that researchers who choose variables can convey their preferences for model size by controlling the tightness of the standard prior of Sims and Zha (1998).

Tables 2 and 3 also report how the findings change as we vary the number of lags. The rank correlation between the posterior probabilities in the baseline with four lags ($P = 4$) and with a different number of lags is very high. We conclude that the ranking of variables is robust to the number of lags. Another result is that the posterior probability of Granger causal priority falls as we add lags (see the row “average GCP probability”). Evidently the interactions between the variables have nontrivial dynamics, and therefore adding lags of the y_j ’s strengthens the predictive power of the y_j ’s for y_i . In particular, monetary aggregates and private sector

TABLE 3.—POSTERIOR PROBABILITY THAT REAL GDP, PRICE LEVEL, AND SHORT-TERM INTEREST RATE ARE GRANGER CAUSALLY PRIOR TO A VARIABLE, ROBUSTNESS, UNITED STATES

Variable	Loosest $\xi = 4$	Looser $\xi = 2$	Baseline $\xi = 1$	Tighter $\xi = 1/2$	Tightest $\xi = 1/4$	Number of Lags, P			
						6	5	3	2
Lending rate to NFCs	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Capacity utilization	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Unemployment rate	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Change in real inventories	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Two-year government bond yield	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Real investment	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01
Price of oil	0.00	0.01	0.00	0.00	0.01	0.01	0.00	0.00	0.00
Ten-year government bond yield	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.03
Industrial confidence	0.21	0.03	0.00	0.00	0.00	0.00	0.00	0.00	0.02
Mortgage interest rate	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.02
Purchasing managers' index	0.01	0.02	0.00	0.01	0.01	0.00	0.00	0.00	0.01
Real imports	0.00	0.00	0.00	0.02	0.01	0.00	0.00	0.00	0.00
Euro-area consumer prices	1.00	0.64	0.00	0.00	0.01	0.01	0.01	0.01	0.09
Producers price index	0.00	0.00	0.00	0.01	0.02	0.00	0.01	0.01	0.05
Real exports	0.00	0.00	0.01	0.01	0.01	0.00	0.00	0.01	0.14
Stock index	1.00	0.29	0.01	0.00	0.01	0.00	0.00	0.08	0.57
Real consumption	0.21	0.06	0.01	0.00	0.00	0.00	0.00	0.03	0.02
Corporate bond spread	0.88	0.17	0.01	0.00	0.00	0.00	0.00	0.01	0.08
Employment	0.02	0.05	0.02	0.03	0.01	0.01	0.02	0.05	0.09
M1	1.00	0.98	0.02	0.00	0.00	0.14	0.08	0.00	0.00
Hours worked	0.03	0.10	0.03	0.01	0.01	0.00	0.01	0.03	0.03
Consumer prices excluding energy, food	1.00	0.99	0.07	0.04	0.07	0.54	0.33	0.00	0.00
Nominal effective exchange rate	0.01	0.26	0.15	0.03	0.01	0.10	0.14	0.20	0.15
Consumer confidence	1.00	0.48	0.15	0.08	0.02	0.02	0.04	0.41	0.70
Commodity prices	0.02	0.54	0.25	0.05	0.02	0.10	0.14	0.56	0.62
Real government consumption	1.00	0.99	0.36	0.07	0.03	0.24	0.29	0.58	0.97
Stock volatility index	1.00	0.79	0.39	0.16	0.05	0.09	0.21	0.47	0.81
Unit labor cost	1.00	0.77	0.46	0.39	0.17	0.14	0.25	0.81	0.92
Loans to NFCs	0.00	0.31	0.48	0.16	0.02	0.33	0.43	0.59	0.89
Euro area real GDP	0.93	0.70	0.53	0.13	0.02	0.07	0.21	0.90	0.95
Real housing investment	0.02	0.24	0.57	0.40	0.02	0.08	0.19	0.92	0.98
Loans for house purchase	0.11	0.56	0.59	0.20	0.03	0.45	0.55	0.69	0.98
Government debt	0.01	0.18	0.64	0.63	0.11	0.10	0.26	0.97	1.00
House prices	0.01	0.35	0.91	0.69	0.19	0.31	0.56	1.00	1.00
Consumer loans	1.00	1.00	1.00	0.97	0.23	1.00	1.00	1.00	1.00
Dollar-euro exchange rate	1.00	1.00	1.00	0.55	0.00	1.00	1.00	1.00	1.00
Eonia	0.99	0.98	1.00	1.00	1.00	1.00	1.00	1.00	1.00
M2	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Average GCP probability	0.38	0.36	0.25	0.18	0.08	0.18	0.20	0.33	0.40
Rank correlation with the baseline	0.68	0.84	1.00	0.95	0.72	0.91	0.95	0.96	0.92
$\log p(Y \omega^*) - \log p(Y \omega^U)$	298	156	75	31	8	82	76	68	55
$\log p(Y \Omega)$	3,239	3,460	3,536	3,507	3,423	3,511	3,523	3,538	3,526

See the note to table 2.

credit aggregates become more useful in the euro area as the number of lags rises.

The last row in tables 2 and 3 shows how the log marginal likelihood conditional on the set of models Ω , $\log p(Y|\Omega)$, varies across the different specifications. Across the ξ 's, we achieve the best fit to the data in the baseline. Across the P 's in the euro area, the fit peaks at $P = 5$, but the fit at $P = 4$ (baseline) is essentially identical. In the United States, the fit peaks at $P = 3$, but again the fit at $P = 4$ is practically identical.

If we return to the penultimate row, we see that $\log p(Y|\omega^*) - \log p(Y|\omega^U)$ is equal to a positive number regardless of the prior. This finding indicates that Granger-noncausality restrictions are useful no matter which prior we consider. For the baseline, prior Granger-noncausality restrictions are very useful. The log Bayes factor in the favor of ω^* relative to ω^U is 127 in the euro area and 75 in the

United States. A log Bayes factor of 75 corresponds to posterior odds of about 10^{32} to 1. Thus, if one is interested in the choice of variables, there is overwhelming support for omitting some variables from the VAR. Of course, it is critical that the methodology determines which variables to omit. Dropping an arbitrary subset of the variables can cause the fit to deteriorate. As an example, we computed the log Bayes factor relative to the unrestricted model in favor of a model with a Granger-noncausality restriction very different from the Granger-noncausality restriction in model ω^* . The log Bayes factor was -141 in the euro area and -218 in the United States, indicating overwhelming evidence in favor of the unrestricted model.

Finally, to investigate the sensitivity of the findings to the sample, we redid the baseline analysis having dropped from the sample the last four quarters, the last eight quarters, and the last twelve quarters. The rank correlation between the

posterior probabilities in the baseline sample and in each of the shorter samples was very high. The details are in appendix I.

VI. Conclusion

We derive a closed-form expression for the posterior probability of Granger noncausality in a gaussian vector autoregression with a conjugate prior. We also express in closed form the posterior probability of Granger causal priority, a more general relation suitable in a multivariate context. One can use these results to answer the classic question of macroeconomics: Is variable z relevant for variable x ? One can also use these results to choose variables for a VAR.

We think of the application as an illustration, certainly not the final word on which macroeconomic and financial variables interact most closely with income, the price level, and the short-term interest rate. We were interested in modeling the euro-area economy and therefore wanted to apply the methodology to euro-area data. We then had to accept the fact that the euro-area data sample is short, as well as the fact that a financial crisis occurs in the sample. In the future, it will be useful to redo this paper's analysis with models that may fit better in this sample than a VAR with constant variances of the residuals. In some models, such as a VAR with stochastic volatility and a VAR with Markov-switching variances of the residuals, the principle behind the choice of variables will be the same as the principle used in this paper. However, the computation of marginal likelihood will be more complex than here.

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